

Pathmika Weeraratna

Associate Tech Lead | Web & Backend Engineer (Next.js, React & REST APIs)

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LinkedIn | GitHub

Summary

Web and backend-focused engineering leader with five years of experience delivering secure, scalable applications for telecom operators, having grown through every engineering level at Omobio into Associate Tech Lead. Contributed to 20+ client and product engagements across delivery, architecture, and pre-sales in South Asia, Africa, and the Middle East. Hands-on expertise in Next.js/React web platforms, REST API design, payment-gateway integration, and large-scale data/reporting systems (Elasticsearch, PostgreSQL, ClickHouse), alongside a track record of mentoring teams and owning delivery from architecture through production. Deepening technical foundations through an MSc in Artificial Intelligence at the University of Moratuwa. Recognized as Emerging Employee of the Year (2023).

Experience

Associate Tech Lead

Jan 2026 - Sep 2026

Omobio (Pvt) Ltd

- Led, mentored, and scaled a cross-functional engineering team of 10+ developers, directing solution design and delivery across multiple product initiatives including AI/ML-integrated and centralized reporting platforms.
- Served as technical subject-matter expert for pre-sales engagements across Africa, South Asia, and the Middle East, producing architecture proposals and technical validation.
- Led the SOR (Steering of Roaming) centralized reporting platform, architecting full-stack telecom data-reporting solutions from backend pipelines through web dashboards.
- Leveraged AI tools throughout the design and development workflow to boost personal productivity, enabling faster turnaround and higher-quality output.
- Managed client engagements end-to-end, from requirements through production delivery.

Senior Software Engineer

Jan 2024 - Jan 2026

Omobio (Pvt) Ltd

- Led Agile delivery of end-to-end web applications, owning architecture design, effort estimation, and SOW reviews.
- Contributed to SQL/CDR reporting engineering for an SMSC billing platform: reverse-engineered field mappings, recovered ~85,000 dropped records, and fixed a ~100x query-performance regression via targeted database indexing.
- Built a fraud-detection REST service (FastAPI) for the SMS Firewall, combining embedding-based semantic similarity search with a local-LLM GenAI layer.
- Led delivery of HOPP, Hutch's Next.js 14 web self-care portal, owning frontend architecture and payment-gateway integration across multiple payment methods, and took it to production.
- Integrated third-party payment-gateway services end-to-end across multiple payment methods, covering authentication, request/response handling, error mapping, and reversal flows.
- Mentored a team of 10 junior developers, conducted code reviews, and established coding standards, testing practices, and GitHub Actions CI/CD pipelines.

Software Engineer

May 2022 - Jan 2024

Omobio (Pvt) Ltd

- Led development of a centralized web-based telecom reporting platform with asynchronous task processing (Celery) and containerized deployment (Docker); stack: Python, PostgreSQL, ClickHouse, Elasticsearch, Apache Superset, Prometheus, Grafana.
- Contributed to the React admin-console frontend of Dialog's MDA device-management platform (built with Redux state management), and delivered targeted frontend bug fixes on Dialog's CDR-disclosure portal.
- Enforced Conventional Commits via Husky + lint-staged + Commitlint git hooks, standardized Git branching strategy company-wide (later delivered as an internal training program), and contributed to ISO 27001 certification readiness.
- Received the Emerging Employee of the Year Award (2023) for outstanding technical contribution.

Software Engineer (Intern) - Full Stack

Oct 2021 - Apr 2022

Omobio (Pvt) Ltd

- Supported frontend and backend feature development for enterprise-grade projects using React.js, Ext.js/Sencha Touch, and RESTful APIs; transitioned to a full-time Software Engineer role based on strong performance.

Key Projects

- **HOPP Web Self-Care Portal, 2024:** Led the project team; developed a Next.js/TypeScript customer self-care portal with payment-gateway integration, delivered under a Hybrid Git Flow model and deployed to production.
- **SMS Firewall AI/ML Suite (Reporting CMS):** Built the Next.js admin CMS surfacing outputs and rule controls for the suite's five detection modules (OTP-bypass, phishing-URL, encoded-Unicode, grey-route, campaign detection), sitting atop the FastAPI detection backend.
- **SOR (Centralized Reporting Platform):** Centralized telecom data-reporting platform spanning backend pipelines through web dashboards, with asynchronous task processing and containerized deployment (see Experience).
- **FirewallBI (Telecom Reporting Platform), 2023-2025:** Architected and led delivery of a centralized on-prem BI platform for telecom reporting: hundreds of Logstash JDBC pipelines, custom Kibana RBAC, a mutual-TLS Elasticsearch cluster, and a Python reconciliation framework.
- **MDA (Dialog Device Management Platform):** Carrier-grade device-management platform (PHP Yii2 backend, Erlang/OTP signaling core). Contributed to the React admin-console frontend, built with Redux state management.
- **SMS Opt-In/Opt-Out Platform (Bangladesh), 2023:** Reviewed the SRS and contributed solution architecture for a language-preference platform: a React/PHP Yii2/Erlang stack serving 5 user roles with low-latency REST APIs.

Additional Experience & Engagements

Beyond web, delivered cross-platform mobile applications in React Native, including a fully custom native alarm engine (~2,700 lines of native Java/Swift) and a RevenueCat-based subscription integration, and owned Hutch Selfcare for 2+ years, leading its RN version upgrade (0.66.5 → 0.76.5) and 3+ feature change requests. Also contributed to further engagements, including Dialog's CMDP, RetailHub, WinBack, and DF platforms (Sri Lanka) and the RoamerSteering ELK analytics platform (2023), plus pre-sales solution proposals for operators across South Asia and Africa.

Areas of Expertise

- **Web:** React.js, Next.js, TypeScript/JavaScript, Redux, Tailwind CSS
- **Backend & APIs:** Python, FastAPI, REST API design & development, Microservices, Java (Spring Boot), .NET (ASP.NET Core), PHP (Yii2)
- **Data Engineering:** PostgreSQL, ClickHouse, MariaDB/MySQL, Elasticsearch/ELK, Apache Superset, Apache Airflow, Celery
- **Payments & Integration:** OAuth2, payment-gateway integration, card-network payment methods, SMPP, REST/webhook integration
- **DevOps:** Microsoft Azure (AZ-900 Certified), Docker, Git, CI/CD (GitHub Actions, GitLab CI, Jenkins), Linux, Prometheus, Grafana
- **Leadership:** Agile delivery, technical mentoring, pre-sales & client engagement, architecture ownership

Education

Master of Science in Artificial Intelligence (Reading) - University of Moratuwa July 2025 - Present (Expected 2027)

B.Sc. in Information Systems, Second Class Upper Division, GPA 3.44 - University of Colombo School of Computing
2019 - 2022

Diploma in English (Certificate Level), Distinction Pass - Seagis Campus, Nugegoda 2018

Certifications

Microsoft Certified: Azure Fundamentals (AZ-900) - [Verify Credential](#) Aug 2026

Achievements

- Emerging Employee of the Year - Omobio (2023): recognized for exceptional job performance and contribution to key software engineering projects.
- Director's List Awardee - University of Colombo (2021): achieved GPA above 3.7 in first academic year.